

A Distributed Beam Hopping Strategy With Load Balancing and Coordinated Interference Avoidance for Heterogeneous Satellite Systems

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With the rapid development of low Earth orbit (LEO) satellite networks, the issues of spectrum resource scarcity and intersatellite interference have become increasingly prominent. Traditional fixed and centralized multibeam satellite systems suffer from a lack of flexibility and single-point failure vulnerability, making them inadequate for handling dynamically changing traffic demands. Beam hopping (BH) technology enables flexible resource allocation; however, due to the uneven distribution of ground traffic demand and its temporal variations, efficiently allocating beams remains a critical challenge. In this article, we propose a distributed load balancing and interference avoidance BH (DLBIA-BH) protocol within a distributed heterogeneous satellite network (DHSN). The core concept of DLBIA-BH is to first implement a load balancing mechanism that distributes network traffic evenly among LEO satellites. Then, each LEO satellite employs a real-time (RT) traffic-supporting BH mechanism. Finally, medium Earth orbit satellites coordinate the BH resource scheduling among LEO satellites to mitigate intersatellite interference and enhance overall network performance. Compared with benchmark methods, DLBIA-BH maximizes the overall network throughput and RT traffic throughput in DHSN while also improving the total traffic and RT traffic satisfaction rate across the network.

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NOMENCLATURE

Parameter Definition

N_{LEO}	Number of LEO satellites.
N_{MEO}	Number of MEO satellites.
N_c	Number of ground cells.
N_b	Number of beams on a single LEO satellite.
N_t	Number of time slots in a BH frame.
$CT_{i,j}$	The minimum connectivity duration threshold for LEO satellite i and ground cell j .
CS_j	Set of connectable LEO satellites for ground cell j .
L_i^u	Capacity utilization ratio of LEO satellite i .
L_i^{ru}	Available capacity ratio of LEO satellite i .
L_j^{tru}	Sum of available capacity ratios of all LEO satellites connectable to ground cell j .
$P_{i,j}^{sl}$	Probability of LEO satellite i being selected by ground cell j .
SL_j	Selected LEO satellite by ground cell j .
L_{df}	Difference between the highest and lowest capacity utilization ratios of all LEO satellites.
d	Diameter of a ground cell.
BHP_i	BH pattern set for LEO satellite i .
$T_{\text{th}}^{\text{RT}}$	Threshold value for RT traffic packet survival time.
$T_{\text{th}}^{\text{NRT}}$	Threshold value for NRT traffic packet survival time.
Γ_f	Initial population of the GA after screening based on survival time.
M_r	Mutation rate of the GA.
N_e	Number of elites in the GA.
G_{max}	Maximum number of generations in the GA.
$F(X)$	Fitness function of the GA.
N_m^{ms}	Number of LEO satellites to be served by MEO satellite m .
LS_m	Set of LEO satellites served by MEO satellite m .
Pri_i	Priority of LEO satellite i .
BH_i^{bp}	Current best BH pattern of LEO satellite i .
BH_m^{fl}	Final BH pattern schedule.

1. INTRODUCTION

Low Earth orbit (LEO) satellite networks positioned at altitudes below 2000 km, provide broadband communications, ground network backup, Internet of Things (IoT) connectivity, vehicular networking, and emergency communications through their low latency, high bandwidth, and geographic flexibility [1]. Unlike terrestrial networks, LEO satellites offer global coverage and disaster resilience, making them crucial for 6 G networks [2]. Major companies like SpaceX, Amazon, and OneWeb are investing in large-scale LEO constellations [3], establishing them as key enablers of seamless global communications. Heterogeneous satellite networks combine LEO, medium Earth orbit (MEO), and geostationary orbit (GEO) satellites, integrating their advantages for comprehensive coverage and services. This architecture flexibly selects appropriate satellite tiers based on service requirements and locations, providing seamless

TABLE I
Classification of Existing BH Solutions

	Homogeneous	Heterogeneous
Centralized Control	[9], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22]	[23], [24], [25], [26], [27], [28], [29]
Distributed Control	[30], [31]	Our Work

global coverage for various applications [4]. To better understand and utilize such dynamic satellite networking capabilities, this article focuses on distributed heterogeneous satellite networks (DHSN), employing distributed control of beam resources.

Rapid satellite network development has led to spectrum scarcity and interference challenges. In LEO systems, reducing interference is complex due to numerous satellites. Fast satellite movement and brief ground station connections complicate coordination. Traditional multibeam satellite systems improve spectrum reuse efficiency but cannot adapt to changing traffic demands [5], [6]. Beam hopping (BH) technology addresses these challenges by enabling satellites to dynamically adjust beam coverage positions and durations based on traffic demands, enhancing flexibility, resource utilization, and performance [7]. The digital video broadcasting (DVB) organization's DVB-S2X standard includes specifications for BH [8], highlighting its importance in future satellite communications.

BH scheduling can be implemented using either centralized or distributed control systems. In a centralized approach, a ground network center gathers relevant information and executes the BH scheduling process. However, this method presents a risk of system failure in the event of center malfunction, a risk that is substantially increased during wartime or natural disasters when ground infrastructure becomes particularly vulnerable to attacks or destruction. Distributed control, where LEO satellites independently execute algorithms, avoids these problems but requires complex interference avoidance mechanisms due to limited information sharing, often yielding suboptimal solutions. Despite the promising outlook for BH technology, its practical application still faces challenges, such as single-point failure in centralized control [9], coexistence of heterogeneous satellite systems [10], highly dynamic traffic adaptation [11], and intersatellite coordination in DHSN.

Existing BH protocols can be categorized into four types based on whether the control system is centralized or distributed, and whether homogeneous or heterogeneous satellite networks are used, as shown in Table I.

Most existing BH solutions adopt centralized control, with the majority considering homogeneous satellite network systems [9], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], and a few considering heterogeneous satellite network systems [24], [25], [26], [27], [28], [29]. To address the limitations of centralized control methods, some researchers have proposed BH methods in homogeneous satellite network systems with distributed control [30], [31]. However, these solutions cannot be applied to heterogeneous satellite network system environments. To address the problems of existing solutions, this article investigates BH

in DHSN, specifically designing BH strategies in satellite systems where LEO and MEO satellites coexist, with the aim of enhancing the performance of this heterogeneous satellite network system. The main contributions and innovations of this article are as follows.

- 1) We propose a distributed load balancing and interference avoidance BH (DLBIA-BH) protocol, which is the first BH protocol supporting real-time (RT) traffic for DHSN. The roulette wheel-based load balancing (RW-LB) algorithm dynamically allocates ground cells based on satellite available capacity, effectively addressing load imbalance issues and improving resource utilization efficiency. This will be detailed in Section IV-A.
- 2) We develop the BH pattern optimization-based genetic algorithm (BHPO-GA) for supporting RT traffic and avoiding intrasatellite interference, alongside MEO coordinated-BH scheduling (MC-BHS) algorithm for avoiding intersatellite interference. This architecture combines LEO satellites' distributed computing capabilities with MEO satellites' regional coordination capabilities. These mechanisms are introduced in Sections IV-B and IV-C.
- 3) Through simulation experiments, we validate the effectiveness of DLBIA-BH, demonstrating significant improvements in load balancing, network throughput, RT traffic throughput, and traffic satisfaction rates. The results prove the algorithm's superior performance under various traffic demands and different satellite constellation scales, providing a feasible resource allocation solution for future large-scale DHSNs. The simulation is presented in Section V.

II. RELATED WORK

Numerous studies have explored satellite system BH technology. In the homogeneous, centralized control category, the LO-BBP algorithm utilizes Lyapunov optimization to address dynamic BH and resource allocation challenges in nongeostationary orbit (NGSO) satellite systems with nonuniform traffic demands [12]. Paulson Eberechukwu N. et al. [13] proposed a new clustering algorithm for cell association and a heuristic-based multidimensional resource allocation approach for BH LEO satellites in SAGIN, achieving improved delay, capacity, and throughput performance. Han et al. [14] proposed a novel rate-splitting multiple access (RSMA)-enabled cluster-based BH LEO satellite IoT system that jointly optimizes precoding vectors, common rate allocation, and cluster-based BH pattern design to mitigate interbeam interference and improve traffic demand matching performance. Zhang et al. [15] applied greedy algorithms with beam distance constraints to reduce interference. The MQDB algorithm tackles resource-demand mismatch in BH satellite systems through composite priority scheduling and adaptive carrier segmentation, specifically addressing challenges

from uneven traffic distribution and terminal capability heterogeneity [16]. Lin et al. [17] proposed a low-complexity return-link BH scheduling framework by decomposing the resource allocation challenges of NGSO satellite constellations into user-satellite association, BH pattern design, and resource allocation subproblems. Hu et al. [18] formulated the BH illumination-plan problem as a partially observable Markov decision process (POMDP) aimed at minimizing transmission delay. Zhang et al. [19] proposed a multiobjective deep reinforcement learning (DRL) scheduler for dynamic BH in DVB-S2X that targets throughput, per-beam fairness, and co-channel interference mitigation. Sharma et al. [20] used DRL to optimize precoded cluster hopping in multibeam GEO systems, tackling dynamic resource allocation under unknown traffic/interference. Hu et al. [21] hybridized DRL with a genetic algorithm to make dynamic BH decisions, it reports higher throughput and better fairness than DRL-only and GA-only baselines under dynamic traffic. Ran et al. [22] modeled joint BH and power allocation as an MDP and applied parameterized RL to optimize throughput and energy efficiency simultaneously. However, these studies did not consider heterogeneous satellite network systems in real-world environments.

In the heterogeneous, centralized control category, Gu et al. [23] studied a cofrequency multibeam LEO/GEO system with GEO as the primary, where a centralized controller coordinated spectrum sharing. They proposed a cooperative method combining independent set grouping, fractional programming power control, and queue-weighted scheduling, which protected GEO SINR, boosted LEO throughput and worst-user rates, maintained fairness under high load, and achieved fast convergence. Zhai et al. [24] studied resource allocation in LEO–GEO heterogeneous networks, combining GA and convex optimization techniques to improve system throughput. Wang et al. [25] proposed an architecture with LEO constellations as the primary system and GEO satellites as the secondary system, using ephemeris information sharing mechanisms to predict and avoid potential interference. Zeng et al. [26] addressed large-scale LEO constellations by decomposing resource allocation into connection allocation and power coordination subproblems, designing allocation schemes approaching theoretical limits. DRL has also been applied to heterogeneous satellite networks. Representative works include a centralized network control center (NCC)-based scheme that combines convex optimization with GA search [27], DRL-IABHA for GEO–LEO coexistence that improves throughput and reduces interference [28], and JMBMF-DRL for joint spectrum sharing that enhances overall throughput and GEO-side SINR under QoS constraints [29]. While effective, these methods did not consider the single point of failure risk in centralized networks.

In the homogeneous, distributed control category, Zhao et al. [30] designed a 2-D load-balancing BH mechanism, encompassing load balancing between beams and between satellites, effectively improving intersatellite load imbalance issues. Ren et al. [31] proposed a cooperative BH scheme based on multiagent reinforcement learning,

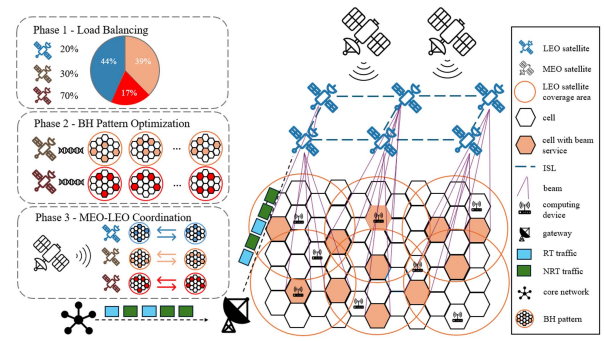


Fig. 1. Example DHSN and the three phases of the proposed DLBIA-BH.

adopting a centralized training and decentralized execution framework that enables adjacent satellites to cooperatively schedule beams, effectively improving throughput. However, such methods are difficult to extend directly to heterogeneous satellite network environments, limiting their scalability.

III. SYSTEM MODEL AND PROBLEM FORMULATION

This section first explains the system model, followed by a description of the problem addressed in this article.

A. Satellite Network Model

This article considers a DHSN composed of LEO and MEO satellites, as shown on the right side of Fig. 1. Without loss of generality, the LEO satellites are configured based on parameters derived from the Starlink satellite constellation to provide communication services to ground cells, while eight MEO satellites are used to coordinate the BH scheduling for LEO satellites. MEO satellites communicate with LEO satellites via broadcasting. LEO satellites communicate with each other through intersatellite links and are equipped with phased array antennas capable of BH functionality. A LEO satellite may be covered by multiple MEO satellites, but it can only be served by one MEO satellite at a time. LEO satellites use beams to provide communication services to ground cells. The Earth's surface is divided into multiple ground cells, each ground cell can be covered by a beam from a LEO satellite. A ground cell may be repeatedly covered by multiple LEO satellites, but at most only one LEO satellite will provide services. Each ground cell is equipped with a computing device that has satellite signal transceiving capabilities, similar to Starlink's smart antenna terminal (such as Dishy McFlatface), and also has computing power that enables it to select which LEO satellite to connect to.

Each LEO satellite uses a frequency reuse method, allowing it to create several beams at once. Each beam can fully use the satellite's bandwidth to cover ground cells that are far enough apart. The propagation channel model is assumed to be under ideal conditions, not including atmospheric attenuation effects, such as rain fading, in order to focus on the beam allocation problem. When a ground cell needs data from the core network, it sends a service request to the LEO satellite it connected to. The

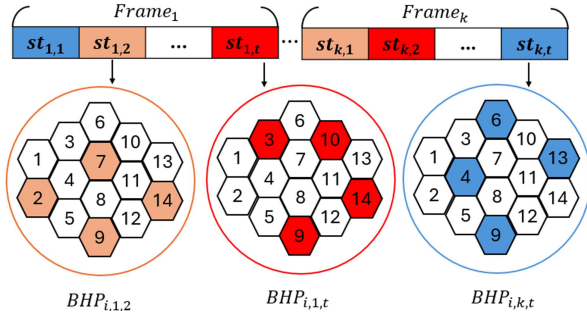


Fig. 2. BH patterns of each LEO satellite.

LEO satellite forwards this request to the gateway, which sends it to the core network. Based on ground cell needs, the core network routes RT and non-RT (NRT) traffic to the gateway, which forwards it to the LEO satellite for final delivery to ground cells via BH techniques. The number of beams that each LEO satellite can use is far less than the number of ground cells to be served. LEO satellite beams adopt time division multiple access, switching to different ground cells at different times. Specifically, time is divided into consecutive frames, with each LEO satellite performing resource allocation once per frame. The frame duration is 640 ms, and since the connection time of a ground cell to any LEO satellite is about 10 min, the connection status of the satellite can be considered unchanged during a frame period. A frame consists of multiple time slots, with a time slot being the unit time a beam serves a ground cell. The time slot t in frame k is denoted as $st_{k,t}$. To achieve optimized resource allocation, each LEO satellite may serve different ground cells in different time slots. The set of ground cells served by all beams of LEO satellite i during time slot t in frame k is termed its BH pattern, denoted as $BHP_{i,k,t}$. As shown in Fig. 2, in each BH pattern, ground cells served by beams are indicated with colored markings. Each LEO satellite in the system may use different BH patterns at different times.

In the DLBIA-BH framework, time is divided into frames. During frame $i - x$, the GA-based BH scheduling computation and control signaling are performed, where x denotes the total overhead in frames. This overhead includes both the scheduling computation time and the transmission delays over the ground-LEO and LEO-MEO control links. An example combined cost is about 648 ms, which corresponds to approximately two frames ($x = 2$).¹

¹Consider a system with 25 LEO satellites, 400 ground cells, and 4 beams per satellite. Encoding 400 cells requires 9 b stored with 2-B alignment. With 4 beams and 64 slots per frame, the payload per LEO is $2 \times 4 \times 64 = 512$ B, or about 520 B including address/control fields. At a control link rate of 13 Mbit/s, this yields a 0.32 ms transmission delay per LEO. For 25 LEOs, the total feedback is 13 000 B (8 ms). Adding propagation delays of about 25 ms (LEO-MEO) and 2 ms (ground-LEO), the total reporting latency is ≈ 35 ms. Together with the GA-based computation time of 613 ms (measured on Intel Core i9-12900K, Ubuntu 22.04 LTS), the combined overhead is ≈ 648 ms ≈ 2 frames. The downlink schedule delivery from the MEO requires 33 ms, which is completed within the same frame i , and each LEO executes the BH schedule in frame $i + 1$.

The signal-to-interference-plus-noise ratio (SINR) for ground cell j served by beam b of LEO satellite i is defined as follows:

$$\text{SINR}_{j,b}^i(t) = \frac{P_b^i |C_{j,b}^i|^2}{\sigma^2 + I_{\text{intra}}^i + I_{\text{inter}}^{\text{LEO}}} \quad (1)$$

where P_b^i is the transmission power allocated by LEO satellite i to beam b , $|C_{j,b}^i|^2$ is the channel gain from beam b of LEO satellite i to ground cell j , σ^2 is the noise power, I_{intra}^i is the intrasatellite interbeam interference of LEO satellite i , and $I_{\text{inter}}^{\text{LEO}}$ is the inter-LEO satellite interference. We simulate transmission quality and interference levels by calculating $\text{SINR}_{j,b}^i$. According to the Shannon's capacity theorem, if ground cell j is illuminated by a beam b of LEO satellite i in time slot t , the available throughput $\lambda_{j,t}$ for that ground cell is $\lambda_{j,t} = B \log_2(1 + \text{SINR}_{j,b}^i)$. The channel gain from the b th beam of LEO satellite i to ground cell j in time slot t is modeled as

$$|C_{j,b}^i|^2 = G_T(\Theta_{j,b}) G_R(\Theta_j) \text{PL}_{i,j} \quad (2)$$

where $G_T(\cdot)$ and $G_R(\cdot)$ denote the satellite transmit and ground receive antenna gains (linear scale), respectively, evaluated at the off-axis angles $\Theta_{j,b}$ (transmit-side, i.e., the satellite beam boresight offset toward cell j) and Θ_j (receive-side, i.e., the ground terminal boresight offset), respectively. The term $\text{PL}_{i,j}$ denotes the linear path gain.

B. Problem Formulation

This article aims to solve the problem of “Designing downlink BH for load balancing and interference avoidance in a DHSN to maximize RT traffic throughput.” The objective of problem P_0 is to maximize RT throughput since RT traffic is delay-sensitive.² To avoid overestimating throughput, we account for both capacity and demand by defining the realized per-cell RT throughput as $\lambda_j^{\text{rt}} = \sum_{t=1}^{N_t} \min(C_{j,t}, D_{j,t}^{\text{rt}})$, where $C_{j,t}$ is the beam capacity (from the Shannon's formula under the current BHP/SINR) and $D_{j,t}^{\text{rt}}$ is the requested RT demand of cell j on slot t . This optimization problem P_0 is defined as follows:

$$\begin{aligned} P_0 : \quad & \max_{BHP_1, BHP_2, \dots, BHP_{N_{\text{LEO}}}} \lambda_{\text{rt}} \\ \text{s.t. } C_1 : \quad & \lambda_{\text{rt}} = \sum_{j=1}^{N_c} \lambda_j^{\text{rt}} \\ C_2 : \quad & \sum_{i=1}^{N_c} BHP_i^t \leq N_b, \quad t = 1, \dots, N_t \\ & i = 1, \dots, N_{\text{LEO}} \\ C_3 : \quad & B_{i,j}^t \in \{0, 1\}, \quad t = 1, \dots, N_t \\ & i = 1, \dots, N_{\text{LEO}}, \quad j = 1, \dots, N_c \end{aligned}$$

²The formulation of P_0 explicitly targets RT traffic as the primary objective, while NRT traffic is incorporated in the BH scheduling stage to enhance resource utilization and is also used as a tie-breaking factor when RT throughput is equal.

$$C_4 : \sum_{b=1}^{N_b} P_b \leq P \quad (3)$$

where N_{LEO} is the number of LEO satellites; N_c is the number of ground cells; N_t is the number of time slots in a BH frame; N_b is the maximum number of beams that can be allocated by a LEO satellite; P_b is the transmission power allocated to beam b ; P is the total power of a each LEO satellite; $\text{BHP}_1, \text{BHP}_2, \dots, \text{BHP}_{N_{\text{LEO}}}$ is defined as the BH pattern set for all LEO satellites in a complete BH frame, each $\text{BHP}_i = \{\text{BHP}_i^1, \text{BHP}_i^2, \dots, \text{BHP}_i^t\}$ includes beam allocation decisions for LEO satellite i across all time slots t , and each $\text{BHP}_i^t = \{B_{i,1}^t, B_{i,2}^t, \dots, B_{i,N_c}^t\}$ includes whether LEO satellite i illuminates ground cell j with a beam in time slot t . Constraint C_1 indicates that λ_{rt} is the sum of throughputs λ_j^{rt} of all ground cells j . Constraint C_2 indicates that each LEO satellite i can allocate at most N_b beams in each time slot t . Constraint C_3 indicates that $B_{i,j}^t$ is a binary variable: 1 if ground cell j served by LEO satellite i is assigned a beam in time slot t , and 0 otherwise. Constraint C_4 indicates that the sum of powers of all beams must be less than or equal to the total power of the LEO satellite.

P_0 is a mixed integer linear programming NP-hard problem [32], which is difficult to find an optimal solution in polynomial time. Therefore, we decompose P_0 into three subproblems: 1) identifying the serving satellite for each ground cell; 2) devising a LEO satellite BH mechanism supporting RT traffic; and 3) mitigating co-frequency interference among LEO satellites.

IV. PROPOSED METHOD

To solve the three subproblems, we propose the DLBIA-BH protocol. Specifically, DLBIA-BH accomplishes three tasks: 1) balancing network load across LEO satellites; 2) designing RT traffic-supported BH for LEO satellites; and 3) avoiding inter-LEO satellite interference. In the following, how DLBIA-BH accomplishes these three tasks is explained in detail. The parameters used in this article are listed in the Nomenclature.

A. Balancing Network Load Across LEO Satellites

To determine the serving satellite for each ground cell and balance the network load, we designed the RW-LB algorithm. This algorithm is executed by each ground cell and employs a probability-based approach for selecting a serving satellite rather than relying solely on the available capacity of the satellites. This method tends to prioritize satellites with lighter network loads for providing services. To prevent connection interruptions, a LEO satellite i is considered connectable to ground cell j if the connectivity duration exceeds $CT_{i,j}$, where $CT_{i,j}$ is the minimum connectivity duration threshold and serves as an adjustable system parameter. These LEO satellites will be added to the set of connectable LEO satellites for ground cell j , denoted as CS_j , and the number of connectable LEO satellites for ground cell j is defined as $|CS_j|$. In RW-LB, the probability

Algorithm 1: RW-LB.

Input: L_i^u .

Output: SL_j .

- 1: **Initialize:** $SL_j = \emptyset$.
 - 2: Obtain CS_j for cell j .
 - 3: **for** $i \leftarrow 1$ to $|CS_j|$ **do**
 - 4: Calculate L_i^{ru} for each LEO satellite i according to L_i^u using (4).
 - 5: **end for**
 - 6: Calculate L_j^{tru} according to (5).
 - 7: Calculate $P_{i,j}^{sl}$ according to (6).
 - 8: Select satellite SL_j using roulette wheel selection based on $P_{i,j}^{sl}$.
 - 9: **return** SL_j .
-

of each satellite being selected is proportional to its available capacity. Let L_i^{cur} and L_i^{max} represent the current and maximum usable capacity of LEO satellite i , respectively, the capacity utilization ratio L_i^u of LEO satellite i is defined as $L_i^u = L_i^{\text{cur}}/L_i^{\text{max}}$. The available capacity ratio L_i^{ru} of LEO satellite is defined as

$$L_i^{ru} = 1 - L_i^u. \quad (4)$$

For each ground cell j with $|CS_j|$ connectable LEO satellites, the total available capacity ratio L_j^{tru} of these $|CS_j|$ LEO satellites can be defined as

$$L_j^{tru} = \sum_{i=1}^{|CS_j|} L_i^{ru}. \quad (5)$$

The probability of LEO satellite i being selected by ground cell j , denoted as $P_{i,j}^{sl}$, can be calculated as

$$P_{i,j}^{sl} = L_i^{ru}/L_j^{tru}. \quad (6)$$

The optimization objective for network load balancing is defined as minimizing the difference L_{df} between the satellite with the highest capacity utilization ($\max_i \{L_i^u\}$) and the satellite with the lowest capacity utilization ($\min_i \{L_i^u\}$), which can be calculated as

$$L_{df} = \max_i \{L_i^u\} - \min_i \{L_i^u\}. \quad (7)$$

The operation of RW-LB is detailed in Algorithm 1. The inputs of RW-LB include the L_i^u of LEO satellites i , $i = 1, \dots, |CS_j|$. The output is the selected LEO satellite SL_j for ground cell j . In RW-LB, the variables are first initialized. Then, ground cell j obtains CS_j that it can connect to and $|CS_j|$ (Lines 1 and 2). Then, according to (4), L_i^{ru} of each connectable LEO satellite i is calculated (Lines 3–5). According to (5), L_j^{tru} of each connectable satellite is calculated, and according to (6), $P_{i,j}^{sl}$ of each connectable satellite being selected is calculated (Lines 6 and 7). Finally, the roulette wheel selection method is used to select the serving LEO satellite SL_j , which is then returned (Lines 8 and 9).

B. Designing RT Traffic-Supported BH for LEO Satellites

To design a LEO satellite BH mechanism that supports RT traffic, we designed the BHPO-GA, which is executed by each LEO satellite. Each satellite optimizes its BH pattern frame by frame. In BHPO-GA, RT traffic has packet survival time thresholds, denoted as T_{th}^{RT} . Expired RT packets will be removed. For each LEO satellite, a set of BH patterns is randomly generated to construct the initial population Γ_f , which serves as the starting point for the GA-based optimization. To aid in describing BHPO-GA, certain notation needs to be defined. Let the diameter of a ground cell be d , the interference level between two beams i and j , denoted as $I_{i,j}$, can be defined as

$$I_{i,j} = \begin{cases} 1, & i \neq j \text{ and } d_{i,j} \leq d \\ 0, & \text{otherwise} \end{cases} \quad (8)$$

where $d_{i,j}$ is the distance between beams i and j , and $I_{i,j} = 1$ indicates the presence of cofrequency interference.

Let X represent an individual in the GA, i.e., a BH pattern in this article. The total intrasatellite interbeam interference, denoted as $\text{Beam}_I(X)$, can be defined as

$$\text{Beam}_I(X) = \sum_{i=1, j=1}^{N_b} I_{i,j}. \quad (9)$$

The fitness function $F(X)$ in BHPO-GA is as follows:

$$F(X) = \alpha \frac{\lambda_{rt}(X)}{\lambda_{rt}^{\max}} + \beta \frac{\lambda_{nrt}(X)}{\lambda_{nrt}^{\max}} - \omega \frac{\text{Beam}_I(X)}{\text{Beam}_{I,\max}} \quad (10)$$

where $\lambda_{rt}(X)$ and $\lambda_{nrt}(X)$ are the throughputs of RT and NRT traffic, respectively, and $\lambda_{rt}^{\max}$ and $\lambda_{nrt}^{\max}$ are the corresponding total traffic demand of RT and NRT traffic, respectively. $\text{Beam}_I(X)$ is the total intrasatellite interbeam interference, and $\text{Beam}_{I,\max}$ is the maximum intrasatellite interbeam interference among all feasible solutions X . The parameters α , β , and ω are nonnegative weighting factors that balance the relative importance of RT throughput, NRT throughput, and interference. The simulation results under different weight settings are shown in Fig. 3. It can be observed that the optimal weights for maximizing the $F(X)$ are $\alpha = 0.7$, $\beta = 0.15$, and $\omega = 0.15$. BHPO-GA uses the multiobjective optimization fitness function $F(X)$ to optimize RT traffic throughput while avoiding intrasatellite interbeam interference and meeting packet survival time requirements. The operation of BHPO-GA encompasses multiple aspects including gene representation, population generation, and evolution. For gene representation, BHPO-GA represents each satellite's beam allocation pattern in a time slot as a set of binary strings. The string length is $\lceil \log_2 N_c \rceil$, with the binary string value representing the ground cell allocated to the beam, and the number of strings in each set determined by the number of beams on the satellite. For example, a satellite with 4 beams facing 35 ground cells requires 6 b to represent ground cell numbers, with a pattern like (001000, 001010, 010001, 010101) representing ground cells numbered 8, 10, 17, and 21 being served by beams in the same time slot. In the initial population generation

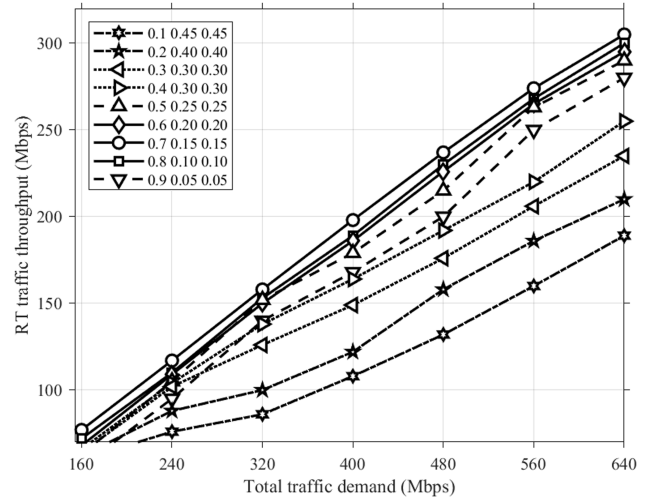


Fig. 3. Impact of different values of α , β , and ω .

phase, BHPO-GA generates an initial population Γ_f composed of multiple individuals γ that meets packet survival time requirements.

For fitness calculation, the algorithm first converts the individual's binary strings to BH patterns, calculates $\lambda_{rt}(X)$ and $\lambda_{nrt}(X)$ in the cells covered by beams, then calculates the interference level between adjacent beams according to (8), and accumulates to get the total interference level $\text{Beam}_I(X)$ according to (9). Finally, it normalizes the fitness function, setting the fitness to the minimum value for individuals that do not meet packet survival time requirements. BHPO-GA implements an elitism strategy, preserving a small number of individuals with the highest fitness in each generation to directly enter the next generation without undergoing crossover and mutation. BHPO-GA uses the roulette wheel method for selection operations, allocating selection probabilities proportionally based on individual fitness values, with higher fitness individuals having a greater chance of being selected. For crossover operations, BHPO-GA uses single-point crossover: randomly selecting parents and crossover points, exchanging genes to produce offspring, while ensuring beam quantity and ground cell constraints are met. To increase population diversity, BHPO-GA also performs mutation operations, randomly selecting positions to flip genes, and checking if the mutated individuals still meet constraints. Finally, the algorithm outputs a set of better BH patterns that meet the requirements.

The operation of BHPO-GA executed by each LEO satellite i is shown in Algorithm 2. The inputs include the initial population Γ_f , the mutation rate M_r , the number of elites N_e , the maximum number of generations $G_{\max}$, and the number of time slots in each BH frame N_t . The output is the optimal BH pattern set BHP_i . The variables are first initialized (Line 1). Then, for each generation g , the fitness value $F(X)$ of each individual γ is calculated. This including calculating λ_{rt} , λ_{nrt} , Beam_I , and $F(X)$ (Lines 2–6). Next, individual selection, crossover, and mutation operations are applied to generate the next generation (Lines 7–12). Specifically, elitism is first applied in Line 7, where

Algorithm 2: BHPO-GA.

Input: $\Gamma_f, M_r, N_e, G_{\max}, N_t$.**Output:** BHP_i .

- 1: **Initialize:** $g = 0$.
 - 2: **for** $g \leftarrow 1$ to $G_{\max}$ **do**
 - 3: **for all** $\gamma \in \Gamma_f$ **do**
 - 4: Calculate $\lambda_{\text{rt}}, \lambda_{\text{nrt}}, \text{Beam}_l$ according to (9).
 - 5: Calculate $F(X)$.
 - 6: **end for**
 - 7: Select N_e elite individuals add to new population.
 - 8: **Selection:** Apply roulette wheel selection based on $F(X)$.
 - 9: **Crossover:**
 Select two parent solutions for single-point crossover ensuring beam count and region constraints are satisfied.
 - 10: **Mutation:**
 Select individuals with probability M_r , flip genes (0 to 1 or 1 to 0), and ensure constraints are satisfied.
 - 11: Update Γ_f with offspring.
 - 12: **end for**
 - 13: Sort Γ_f by $F(X)$ in descending order.
 - 14: Add top N_t individuals from Γ_f to BHP_i .
 - 15: **return** BHP_i .
-

Algorithm 3: MC-BHS.

Input: $N_m^{ms}, N_t, \text{BHP}_i$.**Output:** BH_m^{fl} .

- 1: **Initialize:** $LS_m = \emptyset, \text{BH}_m^{fl} = \emptyset, \text{BH}_{m,t} = \emptyset$.
 - 2: **for** $i \leftarrow 1$ to N_m^{ms} **do**
 - 3: Add i to LS_m .
 - 4: **end for**
 - 5: Calculate Pri_i for each LEO satellite i in LS_m .
 - 6: Sort LEO satellites i in descending order of Pri_i .
 - 7: **for** $t \leftarrow 1$ to N_t **do**
 - 8: **for all** $i \in LS_m$ **do**
 - 9: $\text{BH}_i^{bp} = \text{null}$.
 - 10: **for all** $p \in \text{BHP}_i$ **do**
 - 11: **if** no interference with patterns in $\text{BH}_{m,t}$ **then**
 - 12: $\text{BH}_i^{bp} = p$.
 - 13: **break**.
 - 14: **end if**
 - 15: **end for**
 - 16: **if** $\text{BH}_i^{bp} \neq \text{null}$ **then**
 - 17: Add BH_i^{bp} to $\text{BH}_{m,t}$.
 - 18: **end if**
 - 19: **end for**
 - 20: Add $\text{BH}_{m,t}$ to BH_m^{fl} .
 - 21: **end for**
 - 22: **return** BH_m^{fl} .
-

the top N_e individuals in the current generation are directly retained in the next generation to prevent the loss of high-quality solutions during evolution. After all generations have completed, where individuals are sorted according to their fitness values, and the top N_t individuals are selected as the candidate BH pattern set BHP_i for the corresponding LEO satellite (Lines 13 and 14). Finally, the population Γ_f is updated and BHP_i is returned (Lines 15).

C. Avoiding Inter-LEO Satellite Interference

To prevent interference among LEO satellites, each LEO satellite sends its BH pattern set to its nearest MEO satellite. The MEO satellite subsequently executes the MC-BHS algorithm to determine the BH allocations for LEO satellites, with a focus on prioritizing RT traffic. Specifically, each MEO satellite coordinates the sequence of BH patterns within a frame for the LEO satellites to prevent co-frequency interference among satellites and to enhance the performance of RT traffic. After applying (8) to avoid intersatellite interference, MEO satellites allocate BH patterns in each time slot according to LEO satellites' priority order. It is important to note that MEO satellites do not participate in intersatellite communication for coordination. As a result, LEO satellites situated near the boundaries between different MEO coverage cells may still encounter intersatellite interference, since their BH patterns are not synchronized across MEO satellite boundaries. This represents a necessary compromise to preserve overall system

efficiency, as coordinating BH patterns across MEO satellites would involve increased computational complexity and communication overhead.

The operation of MC-BHS executed by MEO satellite m is shown in Algorithm 3. The inputs include the number of LEO satellites N_m^{ms} that need to be served by MEO satellite m , the number of time slots in a BH frame N_t and the BH pattern set BHP_i of each LEO satellite i served by MEO satellite m . The output is the BH pattern scheduling scheme BH_m^{fl} for LEO satellites served by MEO satellite m . The variables are first initialized (Line 1). Then, MEO satellite m adds each LEO satellite i to its service set LS_m (Lines 2–4). Next, the priority of each LEO satellite i , denoted as Pri_i and determined by the amount of RT traffic, is calculated and sorted in descending order (Lines 5 and 6). For each time slot t in the BH frame, the MEO satellite sequentially processes each LEO satellite i based on priority, selecting from BHP_i the best pattern BH_i^{bp} that causes no interference (Lines 7–15). The quality of patterns is determined by the amount of RT traffic, with the pattern having the most RT traffic being the best pattern. If a BH pattern meeting interference constraints is found, MEO satellite m adds that pattern to the temporary BH pattern set $\text{BH}_{m,t}$ for the current time slot (Lines 16–19). Finally, $\text{BH}_{m,t}$ of the current time slot is added to BH_m^{fl} , which is then returned (Lines 20–22).

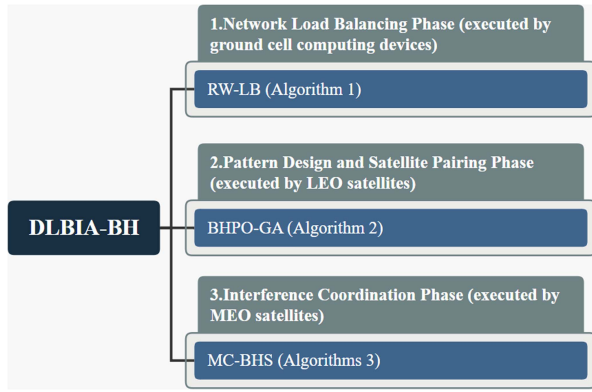


Fig. 4. DLBIA-BH protocol consists of three phases.

TABLE II
Simulation Parameters

parameter	value
Number of LEO satellites	[5, 10, 15, 20, 25]
LEO satellite altitude	550 km
Number of beams	4
Number of MEO satellites	8
MEO satellite altitude	8000 km
Time slot length	10 ms
Number of time slots in a frame during BH	64
Number of ground cells	400
Radius of ground cell	15 km
Longitude range	[75°W, 90°W]
Latitude range	[30°N, 40°N]
Number of simulation cycles	100
Total system bandwidth	150 MHz
Total satellite transmit power	40 dBW
Satellite antenna gain	37 dBi
Ground terminal antenna gain	0 dBi
Noise temperature	290 K

D. Distributed Load Balancing and Interference Avoidance BH

The complete DLBIA-BH protocol combines the methods described in the previous three subsections, as shown in Fig. 4. The proposed DLBIA-BH protocol can be divided into the following three phases, with each phase solving one subproblem. In the network load balancing phase, ground cell computing devices execute Algorithm 1. In the pattern design and satellite pairing phase, LEO satellites execute Algorithm 2. In the interference coordination phase, MEO satellites execute Algorithm 3.

V. SIMULATION RESULTS

We used the MATLAB satellite communications toolbox to build the satellite and ground environment to validate the performance of DLBIA-BH. The simulation parameters are listed in Table II. The simulation environment includes 5 to 25 LEO satellites (with an orbital altitude of 550 km) and 8 MEO satellites (with an orbital altitude of 8000 km). Each LEO satellite is equipped with 4 beams, and the LEO satellite parameters are partially based on the Starlink constellation architecture. The time slot length is set to 10 ms, and each BH frame contains 64 time slots, totaling 640 ms. The simulation area includes 400 ground cells, each with a

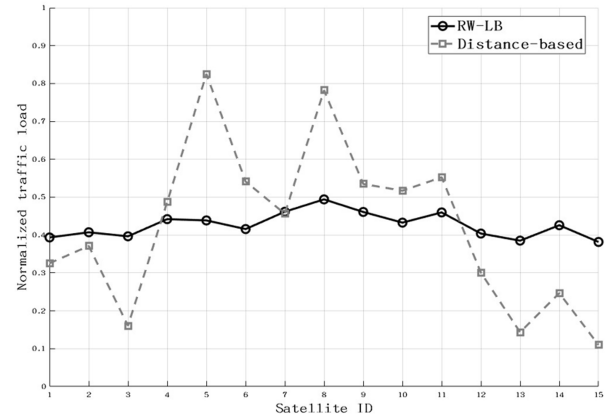


Fig. 5. Performance comparison of load balancing strategies: RW-LB versus Distance-based selection.

TABLE III
Load Disparity Between the Most and Least Loaded Satellites Across Varying Numbers of LEO Satellites

Scheme	Number of satellites				
	5	10	15	20	25
RW-LB	0.01	0.09	0.11	0.09	0.09
Distance-based	0.41	0.77	0.72	0.55	0.47

radius of 15 km, covering part of North America (longitude 75°W to 90°W, latitude 30°N to 40°N). Each ground cell has both RT and NRT traffic demands. The total system bandwidth is 150 MHz, satellite total transmission power is 40 dBW, satellite antenna gain is 37 dBi, ground terminal antenna gain is 0 dBi, and the noise temperature is 290 K. In RW-LB, $CT_{i,j}$ is set to two frames. The simulation results presented as follows are the average of 100 simulation runs.

Balancing network load across LEO satellites is the first task of the DLBIA-BH protocol, and its performance directly affects the effectiveness of subsequent BH optimization and interference coordination. Therefore, the load balance effect of different load balancing strategies is evaluated first. We compare the RW-LB algorithm with a distance-based satellite selection strategy, where a ground cell consistently selects the closest satellite. In the case of a DHSN with 15 LEO satellites, the load balancing degree comparison of two satellite selection strategies is shown in Fig. 5. The RW-LB algorithm achieves a relatively stable distribution with small load differences between satellites, indicating that traffic is evenly distributed among satellites. In contrast, the distance-based satellite selection strategy shows high fluctuation with larger load differences, indicating that some satellites are overloaded while others have idle resources. Specifically, the RW-LB algorithm maintains a load difference of only 0.11 between satellites, while the distance-based strategy exhibits a significantly higher load difference of 0.72. The RW-LB algorithm considers the proportion of available capacity for each satellite, enabling traffic to be dynamically adjusted and distributed according to satellite resources, effectively achieving load balancing.

Table III presents a comparison of load balancing performance between two schemes across different satellite

quantities. The values denote the load disparity, specifically the difference between the satellite with the highest load and the satellite with the lowest load under various satellite configurations. The results demonstrate that the RW-LB algorithm exhibits superior load balancing performance across all satellite numbers. When the number of satellites is small, satellites generally experience overload conditions, where the effectiveness of load balancing is limited. Specifically, when only 5 satellites are deployed, most satellites reach 100% capacity utilization, resulting in a relatively small load disparity of 0.01 for RW-LB since all satellites are operating near their maximum capacity. As the number of satellites increases, load balancing becomes increasingly important. When the satellite count reaches 15 or more, each satellite has sufficient queue resources and no longer experiences overload conditions. The RW-LB algorithm still maintains its advantage, indicating that even under resource-abundant conditions, an effective load balancing algorithm retains significant value.

Next, we compare DLBIA-BH with various existing distributed strategies from the perspective of solving the three subproblems. Since satellites cannot efficiently allocate beams over time to multiple ground cells without a BH algorithm, all methods evaluated must incorporate a solution for the second subproblem. The comparison methods include two methods to assess individual contributions of DLBIA-BH components, plus methods that combine existing BH algorithms with the proposed Algorithms 1 and 3, where existing BH algorithms comprise random, periodic, and greedy approaches. Specifically, we compare the performance of DLBIA-BH with the following five methods.

- 1) *Algorithm 1 combined with Algorithm 2 (denoted as Algo 1+2)*: This method does not handle intersatellite interference.
- 2) *Algorithm 2 combined with Algorithm 3 (denoted as Algo 2+3)*: This method adopts a distance-based satellite selection strategy and does not handle load balancing.
- 3) *Algorithms 1 and 3 combined with random BH (denoted as LBRIA)*: This method differs from DLBIA-BH by allocating LEO satellite beams to randomly chosen ground cells.
- 4) *Algorithms 1 and 3 combined with periodic BH (denoted as LBPIA)*: This method differs from DLBIA-BH by allocating LEO satellite beams to ground cells in a round-robin manner based on ground cell indexes.
- 5) *Algorithms 1 and 3 combined with greedy BH (denoted as LBGIA)*: This method differs from DLBIA-BH by prioritizing the allocation of LEO satellite beams to ground cells with the highest traffic demand.

The traffic generation model uses hotspots and Poisson distributions to simulate realistic network traffic. First, traffic hotspots are established, then the Poisson distribution is utilized to generate the number of packet arrivals for each ground cell in each BH frame based on normalized traffic

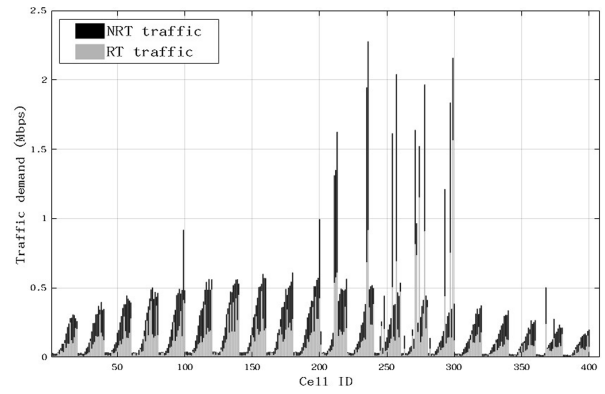


Fig. 6. RT and NRT traffic distribution in different ground cells.

weights and RT/NRT ratios. This model can effectively simulate the traffic differences between densely populated urban cells and suburban regions in satellite networks, as well as the different spatial distribution characteristics of RT and NRT traffic. The distribution of RT and NRT traffic for ground cells in the simulation is shown in Fig. 6.

In the following, the performance of DLBIA-BH is observed in two aspects.

1) Impact of Different Traffic Demands in Ground Cells:

In the network with 15 satellites, Fig. 7 demonstrates the performance of each method under different total traffic demands. Overall, our proposed DLBIA-BH protocol exhibits the best performance across all evaluation metrics, which is attributed to the comprehensive advantages of the DLBIA-BH three-phase coordination mechanism. As illustrated in Fig. 7(a), DLBIA-BH consistently achieves the highest RT traffic throughput across all traffic demands, reaching 304.2 Mbps at 640 Mbps total demand, significantly outperforming other comparison methods which range from 170.7–281.5 Mbps. As shown in Fig. 7(b), DLBIA-BH achieves the highest total throughput of 520.1 Mbps at 640 Mbps demand, demonstrating superior overall system performance compared to other methods ranging from 339.4–472.2 Mbps. More importantly, as shown in Fig. 7(c), at 640-Mbps demand, DLBIA-BH maintains the highest RT traffic satisfaction rate of 96% even under high traffic loads, while other methods show significantly lower performance ranging from 54.8%–92.3%. Similarly, Fig. 7(d) shows that at 640-Mbps demand, DLBIA-BH's overall traffic satisfaction rate of 82.5% maintains a leading position among all methods compared to other methods ranging from 53%–75.2%, proving that DLBIA-BH does not sacrifice overall system performance during RT traffic optimization.

For the two schemes that include partial mechanisms of DLBIA-BH, Algo 1+2 and Algo 2+3, it is observed that the performance of Algo 1+2 is significantly superior to that of Algo 2+3 in most scenarios. As shown in Fig. 7(a), Algo 1+2 achieves 282.3-Mbps RT throughput compared to 235.3 Mbps for Algo 2+3. This result highlights the importance of the load balancing mechanism, which can

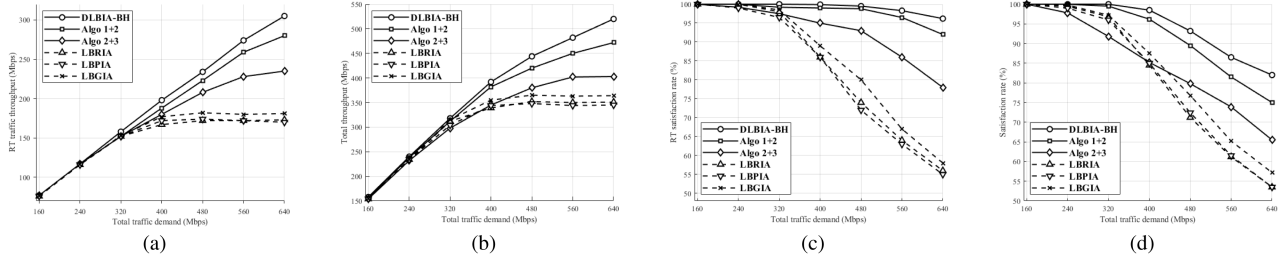


Fig. 7. Performance comparison of different traffic demands. (a) RT traffic throughput. (b) Throughput. (c) RT traffic satisfaction rate. (d) Traffic satisfaction rate.

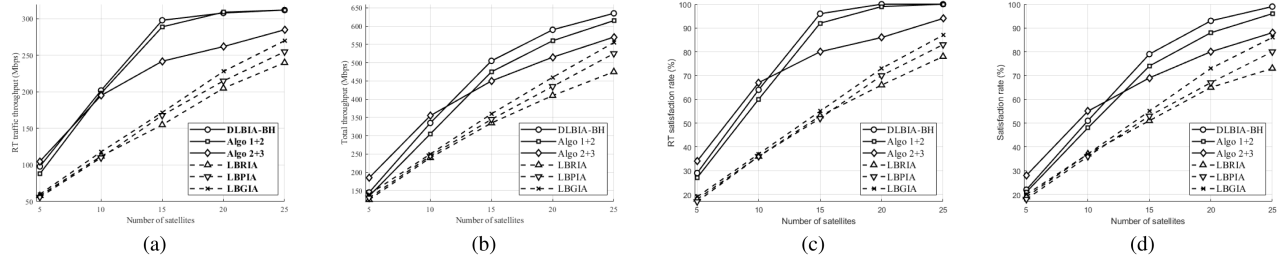


Fig. 8. Performance comparison of varying number of satellites: (a) RT traffic throughput. (b) Throughput. (c) RT traffic satisfaction rate. (d) Traffic satisfaction rate.

effectively distribute traffic evenly among different satellites to prevent a single satellite from becoming a system bottleneck due to excessive load. However, the performance of DLBIA-BH, which incorporates the MC-BHS algorithm, is even more outstanding, proving that considering both load balancing and interference avoidance is necessary to maximize the overall system performance.

The performance of the three methods, LBRIA, LBPIA, and LBGIA, is consistently poor. This is because, despite implementing load balancing and interference avoidance mechanisms, they employ simpler strategies, such as random, periodic, and greedy approaches in the BH pattern design phase. For instance, as shown in Fig. 7(a), their RT throughput ranges only from 169.6–179.1 Mbps. These strategies cannot optimize based on the survival time requirements of RT traffic and intrasatellite interference like the BHPO-GA algorithm in DLBIA-BH. Consequently, their resource allocation efficiency lags significantly, leading to inferior overall performance.

2) *Impact of Varying Number of Satellites:* Under the fixed traffic demand condition of 640 Mbps, Fig. 8 demonstrates the performance of each algorithm under different numbers of LEO satellites. The advantages of DLBIA-BH become more pronounced as the network size increases, demonstrating that DLBIA-BH can maintain high-efficiency operation in large-scale satellite constellations. As shown in Fig. 8(a), with 25 satellites, DLBIA-BH achieves 310.4-Mbps RT throughput, outperforming other methods which range from 236.5–309.9 Mbps. As illustrated in Fig. 8(b), with 25 satellites, DLBIA-BH achieves 633.2-Mbps total throughput, significantly outperforming other methods which range from 474–617.2 Mbps. Fig. 8(c) shows that with 25 satellites, DLBIA-BH achieves 99.9%

RT satisfaction rate, outperforming other methods which range from 76.4%–99.8%. Similarly, Fig. 8(d) demonstrates that with 25 satellites, DLBIA-BH achieves 98.7% overall satisfaction rate, outperforming other methods which range from 74.1%–96.3%. Notably, when the number of satellites is small (e.g., 5), DLBIA-BH's RT traffic throughput of 92.2 Mbps is slightly lower than Algo 2+3's 105.5 Mbps. This occurs because in low-satellite environments, satellites are almost all in an overloaded state, and Algo 2+3, due to using a distance-based satellite selection strategy, achieves better SINR and thus results in superior throughput. However, as the number of satellites increases and load balancing effects manifest, the benefits of DLBIA-BH become more pronounced. With 25 satellites, the performance gap between DLBIA-BH and Algo 1+2 becomes minimal, as both methods share the same load balancing and BH optimization mechanisms. When the number of satellites increases, satellite resources become abundant, making intersatellite interference less critical to overall performance, thus load balancing becomes the dominant performance factor.

For the two schemes that include partial mechanisms of DLBIA-BH, Algo 1+2 and Algo 2+3 show different performance trends. As shown in Fig. 8(a), Algo 1+2's RT performance steadily improves with increasing satellites, with RT throughput rising from 84.1–309.5 Mbps. In contrast, Algo 2+3 shows superior performance with fewer satellites but limited improvement as satellite count increases, with RT throughput ranging from 105.5–289.3 Mbps. This highlights the importance of load balancing mechanisms in large-scale deployments.

The performance of LBRIA, LBPIA, and LBGIA consistently lags across all satellite configurations. For instance, as shown in Fig. 8(a), LBRIA's RT throughput ranges from

57.9–236.5 Mbps, significantly lower than DLBIA-BH. This is due to the limitations in their BH pattern design, resulting in poor performance across all satellite counts.

VI. CONCLUSION

In this article, we proposed the DLBIA-BH protocol in a DHSN environment. DLBIA-BH adopts a three-phase design: first utilizing the RW-LB algorithm to achieve load balancing among LEO satellites, ensuring even distribution of traffic demands; then using BHPO-GA to select and optimize the BH pattern set supporting RT traffic, with special consideration for packet survival time requirements and intrasatellite interbeam interference; finally, using MC-BHS algorithm to coordinate BH scheduling among LEO satellites, avoiding cofrequency interference between satellites. Simulation results demonstrate that the DLBIA-BH protocol effectively achieves network load balancing and substantially enhances both overall and RT traffic throughput. Furthermore, it significantly improves total traffic and RT traffic satisfaction rates, demonstrating comprehensive performance advantages over existing solutions.

While this article addresses BH optimization in DHSN, it does not consider multilayer satellite networks where multiple constellations operate at different altitudes, therefore future research will extend the DLBIA-BH protocol to such multilayer environments for maximizing RT traffic throughput while addressing cross-layer interference avoidance challenges. Specifically, we preliminarily plan to decompose the two-layer BH scheduling problem into three subproblems: 1) deciding whether each ground area should be served by the upper or lower satellite layer; 2) assigning the appropriate serving satellite for each ground area; and 3) scheduling the BH patterns of different satellites. In the future, we aim to develop efficient algorithms to address these subproblems and further tackle the challenges of two-layer BH scheduling.

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